ABOUSSAAD OTHMANE

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Applied Mathematics and Machine Learning graduate with research experience at Imperial College London in dynamical systems, representation learning, filtering and kernel methods, with particular interests in reinforcement learning and generative modelling.

Education

PSL University (#1 in France - QS 2027)

Master in Modelling, Optimisation, Decision and Organisation (MODO) *Sept. 2025 - Oct. 2026*

Research focused Master's on mathematical modelling, optimisation and decision sciences, supported by **Université Paris-Dauphine - PSL**'s Laboratory for Analysing and Modelling Systems for Aiding Decisions (LAMSADE, UMR CNRS7243) and **Mines Paris - PSL**'s Centre for Management Science (CGS, UMR CNRS9217), undertaken in parallel with my final year at CY Tech. Accepted with an Excellence Grant from **École Normale Supérieure - PSL (ENS Ulm)**.

Courses: Graphs and applications, Mathematical Programming, Modelling in Decision Aiding - Operational Research, Preference Modelling and Multicriteria Decision Making, Industrial and Design Models, Decision under Uncertainty, Multiobjective Optimisation, Problem Structuring Methods, Stochastic Programming, Algorithms for Continuous Optimisation.

CY Tech - CY Cergy Paris University (#5 Computer Science School in France - Le Figaro 2026)

Applied Mathematics and Computer Science Engineering Programme - Diplôme d'ingénieur *Sept. 2023 - Oct. 2026*

Relevant Courses: Probability, Statistics, Algorithmics, Databases, Data Mining, Optimisation, Numerical Analysis, Graph Theory, AI, Time Series, PDE.

Result: Top 10% of the class

CPGE Ibn Ghazi, Rabat, Morocco *Sept. 2020 – June 2023*

Preparatory Classes for Grandes Écoles (MPSI–MP*), an intensive two-year programme in mathematics and physics preparing students for highly selective entrance examinations to France's top engineering schools.

Result: Top 10% of the class.

Experience and Projects

Visiting Student, Imperial College London, UK *May 2026 - September 2026*

- Visiting Student at the **Department of Electrical and Electronic Engineering**, Imperial College London, within the Reinforcement Learning & Control Lab of the Intelligent Systems & Networks group, under the supervision of Dr. Yassir Jedra, **Assistant Professor of Machine Learning** at Imperial and former **Postdoctoral Researcher at MIT LIDS**.
- Developed **stable spectral state representations for partially observed nonlinear dynamical systems**, combining **Koopman Theory**, **state estimation**, and **representation learning**, with theoretical guarantees linking source dynamics to stable filtering and prediction. *(Manuscript in preparation for **Learning for Dynamics & Control (L4DC))***

Visiting Student, Imperial College London, UK *April 2025 - August 2025*

- Visiting Student at the Department of Mathematics under the supervision of Dr. Boumediene Hamzi (**Caltech, Imperial College London and The Alan Turing Institute**). Worked on **kernel methods** for high-dimensional diffusion maps and **statistical learning**.
- Developed approaches to **learn data-adaptive kernels**, with analysis of **spectral stability** and **scalability** through approximation techniques. This work resulted in the peer-reviewed paper *Adaptive Kernel Selection for Kernelized Diffusion Maps*, published in **Physica A: Statistical Mechanics and its Applications**, Vol. 700 (2026), Article 131853.

Research Project, CY Cergy Paris University, France *October 2024 - April 2025*

- Academic research project at the **ETIS Lab (Information Processing and Systems Team, UMR CNRS 8085)** under the supervision of Dr. Sonia Yassa (**CY Cergy Paris University**).
- Worked on implementing a **surrogate multi-objective optimisation model** with **probabilistic models** for costly evaluations in IoT (Edge/Fog/Cloud settings). *(Manuscript in preparation)*

Research and Publications

Stable State Representations for Partially Observed Dynamical Systems *Manuscript in preparation, 2026*

- **Authors: Othmane Aboussaad**, Yassir Jedra.
- **Target venue:** Learning for Dynamics & Control (L4DC).
- Develops a theoretical framework connecting nonlinear dynamics to stable state representations, filtering, and prediction under partial observability.
- **Topics:** Representation learning, Koopman operators, partially observed dynamical systems, state estimation, filtering, statistical learning.

Adaptive Kernel Selection for Kernelized Diffusion Maps *Physica A, 2026*

- **Authors: Othmane Aboussaad**, Adam Miraoui, Boumediene Hamzi, Houman Owhadi.
- Published in *Physica A: Statistical Mechanics and its Applications*, Vol. 700, Article 131853.
- **DOI:** 10.1016/j.physa.2026.131853 | **arXiv:** 2604.18402.
- **Topics:** Kernel methods, diffusion maps, spectral learning, dynamical systems, statistical machine learning.

Certifications, Awards and Distinctions

Cambridge Centre for AI in Medicine – ML Summer School 2025 *September 2025*
Summer School organised by Prof. Mihaela van der Schaar (**University of Cambridge**) on AI/ML for healthcare: an intensive 5-day programme covering LLMs, generative models, causality, data-centric AI, time series, uncertainty, digital twins, AutoML, and responsible AI, with clinical applications.

Hack-Nation Global AI Hackathon *August 2025*
Global 24-hour online AI hackathon organised in collaboration with the **MIT Sloan AI Club**, bringing together students and professionals around applied AI challenges.
Contribution: Developed a minimal prototype inspired by *AlphaFold* for protein structure prediction.

Excellence Scholarship from the Graduate Programme in Computer Science - PSL Research University *June 2025*
Awarded by the École Normale Supérieure's (ENS-PSL) Master's Committee for outstanding academic performance to admitted students in the year 2025–2026 of the MODO Master specialisation (Modelling, Optimisation, Decision & Organisation).

2025 SpaceHACK for Sustainability *April 2025*
Two-day hackathon co-organised by **Arizona State University** (Library Unit & Interplanetary Initiative), the **University of Toronto** and **CY Cergy Paris University**.
Selected track: "Environmental Migration in Sahelian Africa" — using satellite data to analyse climate dynamics in the Sahel and propose proactive solutions for communities.

Interdisciplinarity Award, Hi!ckathon 2024 *December 2024*
Awarded among 59 teams (∼320 participants) for *AquaShield*, an interdisciplinary project combining data science, water-resource management, and financial modelling.

Skills

Programming: Python, Julia, R, SQL, C, Java
Machine Learning & Scientific Computing: NumPy, SciPy, PyTorch, Matplotlib
Research: Kernel methods, representation learning, Koopman operators, state estimation and filtering, time series, optimisation, uncertainty quantification
Tools: GitHub, Linux/UNIX, LaTeX, Jupyter
Languages: Arabic (native), French (bilingual), English (fluent), Spanish (intermediate)